Lecture 5 (part 1): Duality Theory

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Materials are taken from different sources with modifications. I do not claim any authorship of these materials.

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Duality Theory

Duality Theory

- The **dual** problem is defined systematically from the **primal** (or original) LP model.

- The two problems are closely related, *in the sense that the optimal solution of one problem automatically provides the optimal solution to the other*. As such, it may be advantageous computationally in some cases to determine the primal solution by solving the dual.

- These ideas are important and form a subject called duality theory.

Motivation—Finding Upper Bounds

We begin with an example:

$$\text{maximize } 4x_1 + x_2 + 3x_3$$

Subjected to

$$x_1 + 4x_2 \leq 1$$
$$3x_1 - x_2 + x_3 \leq 3$$
$$x_1, x_2, x_3 \geq 0$$

Our first observation is that every feasible solution provides a lower bound on the optimal objective function value, ζ^*.
for example:

- the solution $(x_1, x_2, x_3) = (1, 0, 0)$ yields $\zeta^* \geq 4$
- the solution $(x_1, x_2, x_3) = (0, 0, 3)$ yields $\zeta^* \geq 9$

But how good is this bound? Is it close to the optimal value?

To answer,

we need to give upper bounds

Let's find upper bound by multiplying the first constraint by 2 and adding that to 3 times the second constraint:

$$2(x_1 + 4x_2) \leq 2(1)$$
$$3(3x_1 - x_2 + x_3) \leq 3(3)$$
$$\overline{\qquad\qquad\qquad\qquad\qquad\qquad}$$
$$11x_1 + 5x_2 + 3x_3 \leq 11$$

Now, since each variable is nonnegative, we can compare the sum against the objective function and notice that

$$4x_1 + x_2 + 3x_3 \leq 11x_1 + 5x_2 + 3x_3 \leq 11$$

Hence,

$$\zeta^* \leq 11$$

We have localized the search to somewhere between 9 and 11.

To get a better upper bound, we again apply the same upper bounding technique, but we replace the specific numbers we used before with variables and then try to find the values of those variables that give us the best upper bound.

we start by multiplying the two constraints by nonnegative numbers, y_1 and y_2, respectively

$$y_1(x_1 + 4x_2) \leq y_1(1)$$
$$y_2(3x_1 - x_2 + x_3) \leq y_2(3)$$

$$(y_1 + 3y_2)x_1 + (4y_1 - y_2)x_2 + (y_2)x_3 \leq y_1 + 3y_2$$

If we stipulate that each of the coefficients of the x_i's be at least as large as the corresponding coefficient in the objective function,

$$y_1 + 3y_2 \geq 4$$
$$4y_1 - y_2 \geq 1$$
$$y_2 \geq 3$$

then we can compare the objective function against this sum (and its bound):

$$\zeta = 4x_1 + x_2 + 3x_3$$
$$\leq (y_1 + 3y_2)x_1 + (4y_1 - y_2)x_2 + (y_2)x_3$$
$$\leq y_1 + 3y_2$$

We now have an upper bound $y_1 + 3y_2$ which we should minimize.

$$\text{minimize } y_1 + 3y_2$$

Subjected to

$$y_1 + 3y_2 \geq 4$$
$$4y_1 - y_2 \geq 1$$
$$y_2 \geq 3$$
$$y_1 \geq 0$$

This problem is called the dual linear programming problem associated with the given linear programming problem.

Dual Problem

Dual Problem

The following is a summary of how the dual is constructed from the (equation form) primal:

1. A dual variable is assigned to each primal (equation) constraint and a dual constraint is assigned to each primal variable.
2. The right-hand sides of the primal constraints provide the coefficients of the dual objective function.
3. The dual constraint corresponding to a primal variable is constructed by transposing the primal variable *column* into a *row* with
   (i) the primal objective coefficient becoming the dual right-hand side and
   (ii) the remaining constraint coefficients comprising the dual left-hand side coefficients.
4. The sense of optimization, direction of inequalities, and the signs of the variables in the dual are governed by the rules in Table below

Rules for Constructing the Dual Problem

Primal problem objective[a]	Dual problem		
	Objective	*Constraints type*[b]	*Variables sign*
Maximization	Minimization	$\geq$	Unrestricted
Minimization	Maximization	$\leq$	Unrestricted

[a] All primal constraints are equations with nonnegative right-hand sides, and all the variables are nonnegative.

Example

Primal	Primal in equation form	Dual variables
maximize $z = 5x_1 + 12x_2 + 4x_3$ Subjected to $x_1 + 2x_2 + x_3 \leq 10$ $2x_1 - x_2 + 3x_3 = 8$ $x_1, x_2, x_3 \geq 0$	maximize $z = 5x_1 + 12x_2 + 4x_3 + 0x_4$ Subjected to $x_1 + 2x_2 + x_3 + x_4 = 10$ $2x_1 - x_2 + 3x_3 + 0x_4 = 8$ $x_1, x_2, x_3, x_4 \geq 0$	

Dual Problem

$$\text{Minimize } w = 10y_1 + 8y_2$$

subject to

$$y_1 + 2y_2 \geq 5$$
$$2y_1 - y_2 \geq 12$$
$$y_1 + 3y_2 \geq 4$$
$$\left.\begin{matrix} y_1 + 0y_2 \geq 0 \\ y_1, y_2 \text{ unrestricted} \end{matrix}\right\} \Rightarrow y_1 \geq 0, y_2 \text{ unrestricted}$$

Example

Primal	Primal in equation form	Dual variables
minimize $z = 15x_1 + 12x_2$ Subjected to $x_1 + 2x_2 \geq 3$ $2x_1 - 4x_2 \leq 5$ $x_1, x_2 \geq 0$	minimize $z = 15x_1 + 12x_2 + 0x_3 + 0x_4$ Subjected to $x_1 + 2x_2 - x_3 + 0x_4 = 3$ $2x_1 - 4x_2 + 0x_3 + x_4 = 5$ $x_1, x_2, x_3, x_4 \geq 0$	

Dual Problem

$$\text{Maximize } w = 3y_1 + 5y_2$$

subject to

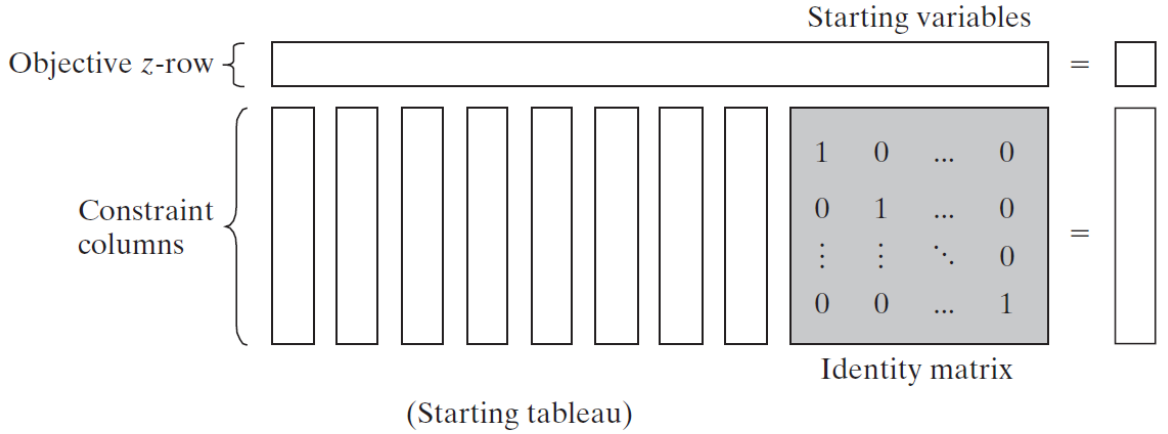
$$\left.\begin{array}{r} y_1 + 2y_2 \leq 15 \\ 2y_1 - 4y_2 \leq 12 \\ -y_1 \leq 0 \\ y_2 \leq 0 \\ y_1, y_2 \text{ unrestricted} \end{array}\right\} \Rightarrow y_1 \geq 0, y_2 \leq 0$$

Primal–dual Relationships

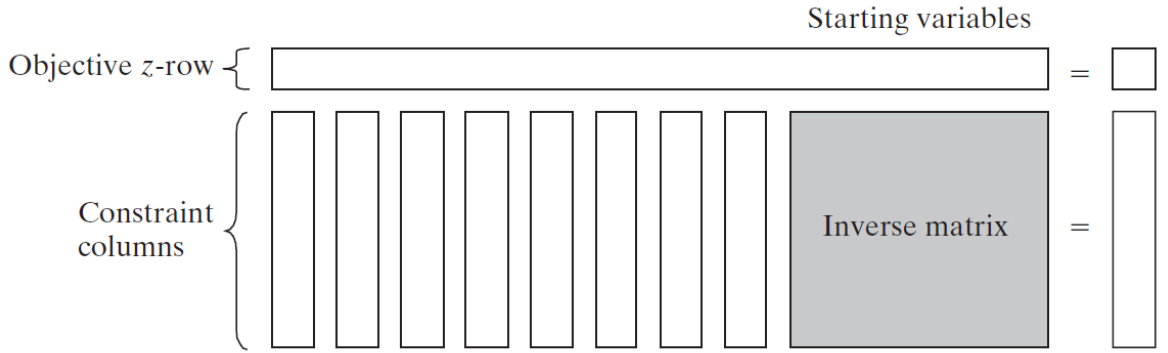
Simplex Tableau Layout

Figure represents the *starting* and *general* simplex tableaus schematically. In the starting tableau, the constraint coefficients under the starting variables form an **identity matrix** (all main-diagonal elements are 1, and all off-diagonal elements are zero).

With this arrangement, subsequent iterations of the simplex tableau generated by the Gauss–Jordan row operations modify the elements of the identity matrix to produce what is known as the **inverse matrix**.



(Starting tableau)



(General iteration)

Optimal Dual Solution

The primal and dual solutions are closely related, in the sense that the optimal solution of either problem directly yields the optimal solution to the other, as is explained subsequently.

Thus, in an LP model in which the number of variables is considerably smaller than the number of constraints, computational savings may be realized by solving the dual because the amount of computations associated with determining the inverse matrix primarily increases with the number of constraints.

Notice that the rule addresses only the amount of computations in each iteration but says nothing about the total number of iterations needed to solve each problem.

We provide two methods for determining the dual values.

Method 1.

$$\begin{pmatrix}\text{Optimal value of}\\ \textit{dual}\text{ variable } y_i\end{pmatrix} = \begin{pmatrix}\text{Optimal primal } z\text{-coefficient of } \textit{starting} \text{ basic variable } x_i\\ +\\ \textit{Original}\text{ objective coefficient of } x_i\end{pmatrix}$$

Method 2.

$$\begin{pmatrix}\text{Optimal values}\\ \text{of } \textit{dual}\text{ variables}\end{pmatrix} = \begin{pmatrix}\text{Row vector of}\\ \textit{original}\text{ objective coefficients}\\ \text{of optimal } \textit{primal}\text{ basic variables}\end{pmatrix} \times \begin{pmatrix}\text{Optimal } \textit{primal}\\ \text{inverse}\end{pmatrix}$$

Optimal Dual Solution

Example

Primal	Primal in equation form	Dual Problem
maximize $z = 5x_1 + 12x_2 + 4x_3$ Subjected to $x_1 + 2x_2 + x_3 \leq 10$ $2x_1 - x_2 + 3x_3 = 8$ $x_1, x_2, x_3 \geq 0$	maximize $z = 5x_1 + 12x_2 + 4x_3 + 0x_4 - MR$ Subjected to $x_1 + 2x_2 + x_3 + x_4 = 10$ $2x_1 - x_2 + 3x_3 + R = 8$ $x_1, x_2, x_3, x_4, R \geq 0$	Minimize $w = 10y_1 + 8y_2$ Subjected to $y_1 + 2y_2 \geq 5$ $2y_1 - y_2 \geq 12$ $y_1 + 3y_2 \geq 4$ $y_1 \geq 0$ $y_2 \geq -M$ y_2 unrestricted

We now show how the optimal dual values are determined using the two methods:

Method 1.

I. In Table, the starting primal variables x_4 and R uniquely correspond to the dual variables y_1 and y_2, respectively.

Optimal Tableau of the Primal

Basic	x_1	x_2	x_3	x_4	R	Sol.
z	0	0	3/5	29/5	$-\frac{2}{5} + M$	$54\frac{4}{5}$
x_2	0	1	-1/5	2/5	-1/5	12/5
x_1	1	0	7/5	1/5	2/5	26/5

Optimal Dual Solution

Example – cntd.

Primal	Primal in equation form	Dual Problem
maximize $z = 5x_1 + 12x_2 + 4x_3$ Subjected to $x_1 + 2x_2 + x_3 \leq 10$ $2x_1 - x_2 + 3x_3 = 8$ $x_1, x_2, x_3 \geq 0$	maximize $z = 5x_1 + 12x_2 + 4x_3 + 0x_4 - MR$ Subjected to $x_1 + 2x_2 + x_3 + x_4 = 10$ $2x_1 - x_2 + 3x_3 + R = 8$ $x_1, x_2, x_3, x_4, R \geq 0$	Minimize $w = 10y_1 + 8y_2$ Subjected to $y_1 + 2y_2 \geq 5$ $2y_1 - y_2 \geq 12$ $y_1 + 3y_2 \geq 4$ $y_1 \geq 0$ $y_2 \geq -M$ y_2 unrestricted

We now show how the optimal dual values are determined using the two methods:

Method 1.

II. Thus, we determine the optimum dual solution as follows:

Starting primal basic variables	x_4	R
Z-equation coefficients	29/5	-2/5+M

Optimal Dual Solution

Example – cntd.

Method 2.

The optimal inverse matrix, highlighted in Table (Optimal Tableau) under the starting variables x_4 and R, is

Optimal Tableau of the Primal

Basic	x_1	x_2	x_3	x_4	R	Sol.
z	0	0	3/5	29/5	$-\frac{2}{5}+M$	$54\frac{4}{5}$
x_2	0	1	-1/5	2/5	-1/5	12/5
x_1	1	0	7/5	1/5	2/5	26/5

$$\text{Optimal inverse} = \begin{pmatrix} \frac{2}{5} & -\frac{1}{5} \\ \frac{1}{5} & \frac{2}{5} \end{pmatrix}$$

The order of the optimal primal basic variables in the Basic-column is x_2 followed by x_1.
The elements of the original objective coefficients for the two variables must appear in the same order—namely,

$$(\text{Original objective coefficients}) = (\text{Coefficient of } x_2, \text{ coefficient of } x_1) = (12, 5)$$

The optimal dual values are

$$(y_1, y_2) = (\text{Original objective coefficients}) \times (\text{Optimial inverse})$$

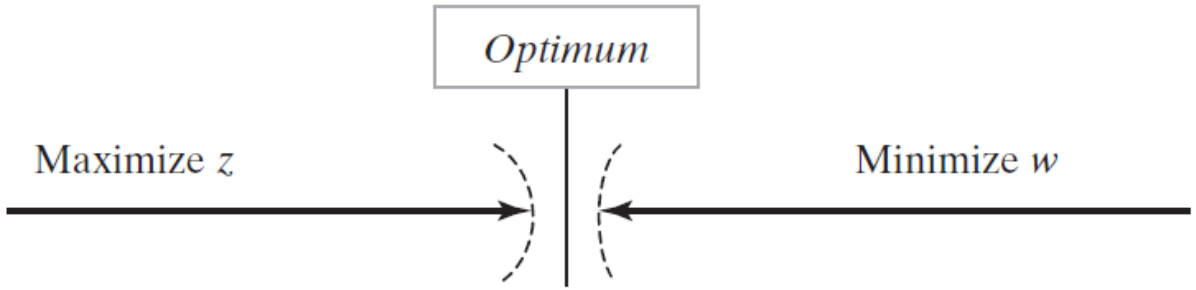
$$= (12,5)\begin{pmatrix} \frac{2}{5} & -\frac{1}{5} \\ \frac{1}{5} & \frac{2}{5} \end{pmatrix} = (\frac{29}{5}, -\frac{2}{5})$$

Primal–dual Objective Values

- For any pair of feasible primal and dual solutions,

$$(\textit{Objective value in the maximization problem}) \leq (\text{Objective value in the minimization problem})$$

- At the optimum, the relationship holds as a strict equation, meaning that the two objective values are equal.
- Note that the relationship does not specify which problem is primal and which is dual.
- Only the sense of optimization (maximization or minimization) is important in this case.
- The optimum cannot occur with z strictly less than w (i.e., $z < w$) because, no matter how close the two values are, there is always room for improvement, which contradicts optimality as Figure demonstrates.



Optimal Dual Solution

Example

Primal	Primal in equation form	Dual Problem
maximize $z = 5x_1 + 12x_2 + 4x_3$ Subjected to $x_1 + 2x_2 + x_3 \leq 10$ $2x_1 - x_2 + 3x_3 = 8$ $x_1, x_2, x_3 \geq 0$	maximize $z = 5x_1 + 12x_2 + 4x_3 - MR$ Subjected to $x_1 + 2x_2 + x_3 + x_4 = 10$ $2x_1 - x_2 + 3x_3 + R = 8$ $x_1, x_2, x_3, x_4, R \geq 0$	Minimize $w = 10y_1 + 8y_2$ Subjected to $y_1 + 2y_2 \geq 5$ $2y_1 - y_2 \geq 12$ $y_1 + 3y_2 \geq 4$ $y_1 \geq 0$ $y_2 \geq -M$ y_2 unrestricted

$\left(x_1 = 0, x_2 = 0, x_3 = \frac{8}{3}\right)$ and $(y_1 = 6, y_2 = 0)$ are (arbitrary) feasible primal and dual solutions.

The associated values of the objective functions are

$$\text{Maximization (primal): } z = 5x_1 + 12x_2 + 4x_3 = 10\frac{2}{3}$$

$$\text{Minimization (dual): } w = 10y_1 + 8y_2 = 60$$

Since $z < w$, the solutions are not optimal.

The optimum value of $z(= 54\frac{4}{5})$ falls within the range $(10\frac{2}{3}, 60)$.

Simplex Tableau Computations

Here, we show how any iteration of the simplex tableau can be generated from the original data of the problem, the inverse associated with the iteration, and the dual problem.

We can divide the computations into two types:

1. Constraint columns (left-hand and right-hand sides).

2. Objective z-row.

Formula 1: Constraint column computations. In any simplex iteration, a left-hand or a right-hand side column is computed as follows:

$$\begin{pmatrix}\text{Constraint column}\\ \text{in iteration } i\end{pmatrix} = \begin{pmatrix}\text{Inverse in}\\ \text{iteration } i\end{pmatrix} \times \begin{pmatrix}\text{Original}\\ \text{constraint column}\end{pmatrix}$$

Formula 2: Objective z-row computations. In any simplex iteration, the objective equation coefficient (reduced cost) of x_j is computed as follows:

$$\begin{pmatrix}\textit{Primal } z\text{-equation}\\ \text{coefficient of variable } x_j\end{pmatrix} = \begin{pmatrix}\text{Left-hand side of}\\ j\text{th } \textit{dual} \text{ constraint}\end{pmatrix} - \begin{pmatrix}\text{Right-hand side of}\\ j\text{th } \textit{dual} \text{ constraint}\end{pmatrix}$$

Simplex Tableau Computations

Example

maximize $z = 5x_1 + 12x_2 + 4x_3$

Subjected to

$$x_1 + 2x_2 + x_3 \leq 10$$
$$2x_1 - x_2 + 3x_3 = 8$$
$$x_1, x_2, x_3 \geq 0$$

Optimal Tableau of the Primal

Basic	x_1	x_2	x_3	x_4	R	Solution
z	0	0	$\frac{3}{5}$	$\frac{29}{5}$	$-\frac{2}{5} + M$	$54\frac{4}{5}$
x_2	0	1	$-\frac{1}{5}$	$\frac{2}{5}$	$-\frac{1}{5}$	$\frac{12}{5}$
x_1	1	0	$\frac{7}{5}$	$\frac{1}{5}$	$\frac{2}{5}$	$\frac{26}{5}$

$$\text{Optimal inverse} = \begin{pmatrix} \frac{2}{5} & -\frac{1}{5} \\ \frac{1}{5} & \frac{2}{5} \end{pmatrix}$$

$$\begin{pmatrix} x_1\text{-column in} \\ \text{optimal iteration} \end{pmatrix} = \begin{pmatrix} \text{Inverse in} \\ \text{optimal iteration} \end{pmatrix} \times \begin{pmatrix} \text{original} \\ x_1\text{-column} \end{pmatrix}$$

$$= \begin{pmatrix} \frac{2}{5} & -\frac{1}{5} \\ \frac{1}{5} & \frac{2}{5} \end{pmatrix} \times \begin{pmatrix} 1 \\ 2 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

Similar computations generate the optimal columns for x_2, x_3, x_4, R, and the right-hand side

Simplex Tableau Computations

Example

maximize $z = 5x_1 + 12x_2 + 4x_3$

Subjected to

$$x_1 + 2x_2 + x_3 \leq 10$$
$$2x_1 - x_2 + 3x_3 = 8$$
$$x_1, x_2, x_3 \geq 0$$

Optimal Tableau of the Primal

Basic	x_1	x_2	x_3	x_4	R	Solution
z	0	0	$\frac{3}{5}$	$\frac{29}{5}$	$-\frac{2}{5} + M$	$54\frac{4}{5}$
x_2	0	1	$-\frac{1}{5}$	$\frac{2}{5}$	$-\frac{1}{5}$	$\frac{12}{5}$
x_1	1	0	$\frac{7}{5}$	$\frac{1}{5}$	$\frac{2}{5}$	$\frac{26}{5}$

Dual Problem

Minimize $w = 10y_1 + 8y_2$

Subjected to

$$y_1 + 2y_2 \geq 5$$
$$2y_1 - y_2 \geq 12$$
$$y_1 + 3y_2 \geq 4$$
$$y_1 \geq 0$$
$$y_2 \geq -M$$
y_2 unrestricted

Next, we demonstrate how the objective row computations are carried out with Formula 2.

The optimal values of the dual variables, $y_1 = \frac{29}{5}, y_2 = -\frac{2}{5}$

Lets compute all the z-coefficients, as illustrated here for x_1 and R.

$$z\text{-cofficient of } x_1 = y_1 + 2y_2 - 5 = \frac{29}{5} + 2 \times -\frac{2}{5} - 5 = 0$$
$$z\text{-cofficient of } R = y_2 - (-M) = -\frac{2}{5} - (-M) = -\frac{2}{5} + M$$

Similar computations can be used to determine the z-coefficients of x_2, x_3, and x_4